

ETERNITY READY RADIO

AzuraCast Enterprise Master Plan

Version 2 · June 2026 · Incorporating PR1–PR13 + AI Features + v5 Roadmap
On-Air Correctness · Enterprise Analytics · Playlist Intelligence · Clock Wheels 10/10 · Quality of Life

Compiled: June 17, 2026 | Repo: github.com/eternityready2/AzuraCast | v0.28.2

What This Document Is

This plan covers everything: what is already built and deployed, what the freelancer v5 roadmap specifies next, enterprise enhancements beyond v5, and what to skip entirely. Built from a full code review of the repo, 16 UI screenshots, the v5 freelancer PDF, and the AzuraWheel Enterprise reference. Nothing already built is listed as new.

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The following features are fully implemented in the codebase at github.com/eternityready2/AzuraCast. This section exists so no completed work gets accidentally re-specced or rebuilt.

Clock Wheel System — PR1 through PR13 (All Complete)

Feature	Status	Notes
PR1 — Timed slot schema	Done	position_seconds (SMALLINT 0–3599) on station_clock_wheel_slots. Migration + entity + API. slot_order as tiebreaker.
PR2 — Schedule conflict detection	Done	ScheduleConflictChecker service. Full/partial/overnight/recurring overlap detection. Runtime skip when non-wheel schedule is active.
PR3 — Hour timeline engine	Done	ClockWheelPlaybackPlanner. Fit-to-window selection. No hard cuts. AutoDJ fallback when no media resolves. playlist_id candidate filter.
PR4 — Validation and logging	Done	position_seconds 0-3599 validation. Structured logging: clock_wheel_id, position_seconds, T, available_seconds, media_id, fallback_reason.
PR5 — Frontend slot editor	Done	EditModal.vue + Entries.vue. mm:ss position input. Hour timeline bar. Overlap/gap warnings. Fill strategy, separation toggle.
PR6 — Full test suite	Done	44+ unit and functional tests. ClockWheelPlaybackPlannerTest, ClockWheelSchedulerTest, SeparationRulesCheckerTest, ClockWheelInheritanceServiceTest, functional Cest.
PR7 — Live Clock Wheel Tab	Done	Schedule page new tab. Rotating CSS clock face in station timezone. Now playing with track title/artist. Current hour slots with queued track names. Conflict warnings. Next hour preview.
PR8 — Liquidsoap enforcement	Done	ClockWheelAnnotator. clock_wheel_enforce_cap flag. cue_out applied via annotation path. Graceful fade — never hard cut.
PR9 — Separation rules + burn rate	Done	SeparationRulesChecker. Artist/title/category separation with configurable windows. Burn rate deprioritization. Relaxation cascade. Template default rules. Daypart overrides.
PR10 — Daypart clock inheritance	Done	ClockTemplate, StationClockWheelTemplate, StationClockDaypart entities. Inheritance service. Template changes propagate to all non-overridden instances. Re-sync button. Schedule time auto-fill.
PR11 — Audit event database	Done	clock_wheel_events table. ClockWheelEventLogger. Stores: timestamp, wheel_id, slot_id, media_id, expected_time, actual_time, drift_seconds, fallback_reason, separation_relaxed, burn_rate_warning.
PR12 — Preview simulator + analytics	Done	ClockWheelPreviewSimulator. GET preview endpoint. AnalyticsAction. AnalyticsModal.vue: tracks queued, fallbacks, avg drift, separation/burn counts, fallback reasons ranked.
PR13 — Emergency override	Done	is_emergency BOOLEAN on StationSchedule. Immediately preempts active clock wheel. Logged as emergency_override in clock_wheel_events. Checkbox in CreateEventModal.

Clock Wheel — Algorithms, Types, and Settings

- Slot algorithms: Random, Oldest Album, Oldest Artist, Oldest Track, Most Recent Album, Most Recent Artist
- Slot types: Music, Talk, ID, Promo, Ad
- Fill strategies: Conservative (defer tight windows) and Aggressive (pick shortest fitting track)
- Separation settings: artist_minutes (default 45), title_minutes (default 90), burn_rate_max_plays_24h

- Per-slot category_id enforcement — same category blocked within artist separation window
- Schedule modes: Flexible and Strict per clock wheel schedule entry
- Slot order tiebreaker when two slots share the same position_seconds

AI Newscaster (Fully Deployed)

Feature	Status	Notes
AiNewsGenerator service	Done	Piper TTS. Voice models: Lessac (default), Joe, Ryan. Produces MP3 news bulletins.
News source scraping	Done	RSS and HTML scraping. Defaults: worthynews.com, raptureready.com. Fully configurable source URLs.
Top/bottom of hour scheduling	Done	Toggle per station: fire at :00, :30, or both. Active hours/days configurable.
Configurable fields	Done	Reporter name, intro script, outro script, story count (max 25), active hours range, active days.
REST API	Done	AiNews GetAction, PutAction, TestPostAction, BulletinGetAction. Full settings CRUD + manual test trigger.
Vue UI	Done	AiNews.vue — full settings form, dashboard with last generation status, bulletin preview, next scheduled time.
CLI command	Done	GenerateAiNewsCommand — manual generation from command line.
UTF-8 fix	Done	Sanitizes scraped headlines before JSON encode. Prevents bulletin generation crashes on special characters.

AI DJ Feature (feature/ai-dj branch — Built, Pending Merge)

Feature	Status	Notes
AiDj entity	Done	name, voice_model_path, shift_intro_template, shift_outro_template, is_enabled. Per-station.
AiDjContent entity	Done	Song intro templates with template variable replacement ({title}, {artist}, etc.). Many-to-many with AiDj.
AiDjSchedule entity	Done	Scheduling for when AI DJ is active. Full recurrence support.
AiDjGenerator service	Done	Piper TTS + ffmpeg. Generates: song intros, shift intros (sign-on), shift outros (sign-off). 400MB disk limit enforced.
AiDjQueueListener	Done	Hooks into AutoDJ queue building. Injects AI DJ audio before queued songs.
AiDjCleanup task	Done	Hourly cleanup of generated audio files. Respects 400MB limit.
REST API	Done	Full CRUD for AiDj, AiDjContent, AiDjSchedule. Test generation endpoint (GET).
Vue UI	Done	AiDj.vue (main settings), AiDjContentLibrary.vue (template library), AiDjScheduleModal.vue.
Unit tests	Done	AiDjQueueListenerTest, AiDjRepositoryTest, AiDjScheduleRepositoryTest, AiDjSchedulerTest.

Other Completed Items

- legalID media type added to StationMedia entity — classifies station identification recordings
- Schedule page upgrade: segmented time input chips replacing raw text inputs
- Clock wheel Re-sync button: force-propagates template changes to all daypart instances
- EditModal sync-time function: auto-fills schedule times from daypart definition

- Version bumped to 0.28.2 with hotfixes for UTF-8 crash and migration naming
- Version 0.28.0: implemented PR7, PR9 remaining items, and PR10 additional features
- Functional test infrastructure wired for clock wheel API (Api_Stations_ClockWheelsCest.php)

Build A1 and A2 together — they are one logical unit

A1 makes songs finish cleanly before :00. A2 makes the station ID fire at exactly :00. Together they define the top-of-hour sound that separates a professional station from a streaming playlist. Build these before any analytics work.

A1 — End-of-Hour Lookahead ★ CRITICAL

The problem

ClockWheelPlaybackPlanner picks music without looking at the hour boundary. A 4-minute song starting at 58:30 runs to 62:30 — past :00. The station ID fires late. Every slot after drifts behind. This is the root cause of the "doesn't go on time" problem.

- Backward pass in ClockWheelPlaybackPlanner before selecting the final music slot of each hour
- If track_end > 3600 (hour boundary), reject that track and try the next shortest candidate in the same category
- Per-wheel setting: "End-of-Hour Window" — seconds before :00 the last song must finish (default 15s, range 0–30s)
- If no track fits the window, ClockWheelAnnotator applies a graceful cue_out fade — never a hard cut
- Log in clock_wheel_events when end-of-hour lookahead activates (new fallback reason type)
- Setting location: Clock Wheel edit form → End of Hour section → ON/OFF toggle + tolerance slider. Default: ON.

A2 — Legal / Top-of-Hour Station ID ★ CRITICAL

Every professional radio station plays a station ID at :00

This is what makes Eternity Ready Radio sound like a real station rather than a streaming playlist. Your Song Playback Timeline already shows you have the IDs: stationid-past-original, station i.d. 1 seth, station i.d. 2 adult, eternity ready radio legal id. Load these into the legal_id playlist immediately when A2 is built.

- Add legal_id as a new case in ClockWheelSlotTypes — separate from the generic id type
- legal_id behavior: NEVER skippable, NEVER deferred by separation rules, ALWAYS fires regardless of burn rate
- Position locked at position_seconds = 0 in every clock wheel — the planner enforces this
- Station setting: Station → Clock Wheels → "Require Station ID at top of hour" ON/OFF toggle
- When ON: every clock wheel template automatically gets a legal_id slot at position 0 if one does not already exist
- Dedicated ID pool: select a playlist of 3–5 short recordings (5–15 seconds each). IDs rotate so same one does not play every hour
- If the legal_id pool is empty: fall back to next id-type track, log a warning, never silence
- Record actual_play_at when the slot fires — stored in clock_wheel_events for drift visibility
- Bottom of hour (:30) support: same logic, separate ON/OFF toggle, separate ID pool

Top-of-Hour Transition — What It Should Sound Like

- Final music song fades out cleanly before :59:45 — end-of-hour lookahead (A1) enforces this
- 0–2 seconds natural pause or crossfade
- Station ID fires at exactly :00:00 — "Eternity Ready Radio" — clean, confident
- First music slot of the new hour begins at :00:10 to :00:20 depending on ID length
- This is the top-of-hour sound that defines a professional station. A1 and A2 together make it happen.

A3 — Category Enforcement UI

The backend already enforces category and playlist per slot. The frontend has no dropdowns — program directors must use the API directly to pin a category to a slot.

- Category dropdown on each slot row in `Entries.vue` and `TemplateEditModal.vue`
- Optional Playlist dropdown on each slot row (overrides category when set)
- Music slots: soft warning badge on slot row if neither category nor playlist is pinned
- No backend changes needed — the enforcement is already wired in `ClockWheelPlaybackPlanner`
- Pool validation on save: warn if assigned playlist has fewer than X tracks matching the slot type

Important: B1–B9 require zero new database tables

All data already exists in `song_history`, `listener`, `analytics`, and `clock_wheel_events` tables. These are new queries and new Vue components on top of existing data. Much faster to build than it looks.

What Already Exists — Do Not Rebuild

- Station Statistics: Best & Worst songs (listener delta), Most Played songs, date range picker
- Listeners by Time Period: daily line chart, hourly bar chart with day-of-week filter, avg vs unique toggle
- Listeners by Day of Week: pie chart showing day distribution
- Listening Time: session length buckets (under 30s through 2+ hours)
- Streams / Browsers / Countries / Clients: full audience breakdown tabs
- Song Playback Timeline: per-song listener count, change delta, source playlist, CSV export
- Live Listener Map: real-time geo pins, unique listeners count, total listener hours
- Listener History: full session log with IP, time, user agent, stream, location, CSV export
- SoundExchange royalty report: existing regulatory export
- Clock Wheel Analytics Modal: per-wheel tracks queued, fallbacks, avg drift, separation/burn counts

B1 — Listener Heatmap Tab [HIGH · Small]

7x24 grid Mon–Sun x 0–23h. Each cell = average listeners for that hour/day. Color scale: light grey = low, deep green = peak. Click any cell to drill into that specific hour across the date range. Zero new backend work — data already returned by the existing `/reports/overview/charts` endpoint. This is the first thing to check when deciding where to schedule your best content.

B2 — Clock Wheel Performance Tab (station-wide) [HIGH · Small]

New tab on main Station Statistics page showing ALL wheels together — not just per-wheel modal. Overall clock accuracy % (slots firing within 10 seconds of target across all wheels). AutoDJ fallbacks ranked by reason: no candidates, separation exhausted, burn rate hit, empty pool. Average drift trend chart — mean drift_seconds per day over the selected date range. Separation relaxation count. Which wheel has the most fallbacks — ranked, click to open that wheel's modal. One new PHP action aggregating existing `clock_wheel_events` data station-wide. No new tables.

B3 — Playlist Performance Tab [HIGH · Small]

New tab in Station Statistics. Plays per playlist in the selected date range — bar chart ranked by volume. Rotation equity per playlist — plays per individual song, shows which songs dominate and which never air. Never-played flag: songs with zero plays in 30 days highlighted in red. Includes B6 playlist-to-listener correlation: average listeners when each playlist was active (`song_history` already stores both `playlist_id` and `listeners_start` on every row). Bar chart comparing average listeners per playlist — which playlists hold the audience best. One new PHP action grouping `song_history` by `playlist_id`. No new tables.

B4 — Song Dropout Tab [HIGH · Small]

Listeners who left within 30 seconds of a song starting. This is a different signal from general tune-out — the song itself is the reason they left. JOIN `song_history.timestamp_start` with `listener.timestamp_end` where listener left within 30 seconds of song start. Songs ranked by dropout count and dropout percentage (dropouts / total plays). Amber flag: dropout rate above 15%. Red flag: above 25%. Particularly valuable for Christian format — identifies songs your audience consistently tunes out on. One new PHP action `ByDropout.php`. No new tables.

B5 — Bot Filtering [HIGH · Small]

Your listener history shows "Generic Bot (Generic Crawler)" from Germany and your own Liquidsoap process as listeners. These inflate Unique Listeners count and Total Listener Hours. Your real human audience number is lower than what AzuraCast reports. Bot detection filter flagging rows where device_is_bot = 1. Separate display: "Real Listeners" vs "Total Connections" with bots excluded from headline number. Liquidsoap connections identified by user agent string (Liquidsoap/x.x.x) and excluded. Settings toggle: "Exclude bots from listener counts" — default ON. Small backend change to AnalyticsRepository and ListenerRepository. No new tables.

B7 — Repeat Listener Tracking [MEDIUM · Small]

listener_hash is already calculated and stored per session. This feature uses that hash to identify returning listeners. New metric: "Returning Listeners" — listener_hashes appearing on more than one calendar day in the period. Loyalty breakdown: listened once / 2–5 times / 6+ times. Average days between return visits for repeat listeners. New listener rate: what % of sessions are first-time listener_hashes. No new tables — computed from existing listener.listener_hash and timestamp_start data. Repeat listener rate is the single most important loyalty metric for internet radio.

B8 — Time-of-Day Growth Trend [MEDIUM · Small]

The existing hourly chart shows average listeners per hour. It does not show whether a specific hour is growing or shrinking week over week. Enhancement to Listeners by Time Period tab — add a "Growth View" toggle. When toggled: instead of average per hour, show week-over-week change per hour. Green cells = growing. Red cells = shrinking. Magnitude shown as percentage change. Uses existing analytics table hourly data — split by week within the selected date range. No new tables — new calculation on existing data in ChartsAction.php.

B9 — Listener Retention Curve [MEDIUM · Medium]

The Listening Time tab shows session length buckets. It does not show WHEN in a session listeners leave. New visualization in the Listening Time tab: retention curve chart. X axis = minutes into session (0–120). Y axis = percentage of listeners still connected at that point. Steep early drop = tune-out at first impression. Cliff at 20 minutes = specific content event causing exits. Overlay: mark on the curve where song transitions happen on average — correlates drop-off with content events. Computed from listener.timestamp_start and timestamp_end. No new tables.

B10 — Daypart Audience Report Tab [MEDIUM · Medium]

The hourly chart shows listeners by hour. What is missing is the view mapped to your named dayparts. New tab in Station Statistics: "Daypart Audience". For each StationClockDaypart, aggregate the matching analytics hourly rows to get average listeners per daypart. Bar chart: average listeners per daypart side by side. Trend view: each daypart's audience over the selected date range — growing or shrinking. Peak and average listeners per daypart shown as summary cards. CSV export. One new PHP action ByDaypart.php joining analytics table with daypart time windows. No new tables.

Enterprise Analytics Additions (Beyond v5)

Song Tune-Out Rate Visualization

Data already captured in Song Playback Timeline (listener + change columns per song). Add a chart layer: visualize listener count before, during, and after each song. Color-code songs: green = gained listeners, red = lost listeners. Sortable by biggest tune-out to identify problem content immediately. Show as an additional view on the Best & Worst tab alongside existing data.

Song Burnout Prediction

Track each song's listener delta trend over its play history. Before it starts tanking, flag it as "approaching burnout" so you pull it from heavy rotation proactively. Calculate a rolling 30-day trend line for each song's listener delta per play. Flag songs whose trend is declining even if individual plays still look neutral. Surface burnout-risk songs in a dedicated report tab.

Clock Wheel Accuracy Grade per Wheel

Single A–F letter grade shown on the clock wheels list page next to each wheel name. Calculated from: % of slots firing within tolerance + fallback rate + separation compliance rate. Makes overall wheel health visible at a glance without opening the analytics modal. Trend arrow showing if this week is better or worse than last week.

Per-Slot Clock Wheel Breakdown in Analytics Modal

Expand the existing analytics modal: table showing each slot's avg drift, fallback count, most-played track, separation relaxation count. Immediately shows which specific slot is the problem. Link slot name to the slot editor for one-click fixing.

Live Listener Count on Clock Wheel Live Tab

Current listener count displayed on Schedule > Live Clock Wheel tab. The now-playing API already returns listener count — this is a frontend-only addition. Update on the same 10-second polling interval already used by the live tab.

Playlist Timeline Filter

Add a playlist filter dropdown to the Song Playback Timeline. Isolate one playlist's performance over time. Identify which playlist drives tune-outs vs holds listeners.

Weekly Auto-Report Email

Auto-generated PDF summary: top songs, listener trends, clock wheel accuracy, fallback reasons. Scheduled delivery — weekly or monthly to station management email without logging in.

CSV Export Upgrades

Add CSV export to: Best/Worst tab, clock wheel analytics modal, new Playlist Performance tab. Currently only Song Playback Timeline and Listeners pages have CSV export.

C1 — Positive Rotation Goals [MEDIUM · Medium]

Problem: a 200-song Gold playlist can cycle the same 20 songs all week. There is no floor. Songs can go unplayed for weeks. New field on StationPlaylist: `rotation_goal_days` (integer, nullable — null means off). In planner candidate filtering: elevate songs that have not played within `rotation_goal_days`. Never-played songs get highest priority within their category — cannot be ignored indefinitely. Log in `clock_wheel_events` when a rotation goal promotion occurs. Surface overdue songs in the B3 Playlist Performance tab. Playlist edit form: new "Rotation Goal" field — "Every song plays at least once every N days". Recommended for Eternity Ready: Power Current = 3 days, Gold = 7 days, Specialty = 14 days. Looking at your Most Played: "stones master (aaron hymes)" at 94 plays and "a lil Church" at 86 plays are dominating. Rotation goals will fix this.

C2 — Per-Slot Separation Overrides [MEDIUM · Small]

Separation rules currently apply at wheel and daypart level. This adds a third level: the individual slot. Your Power Current slot can have tighter artist separation than your Gold slot without changing the whole wheel. New columns on `station_clock_wheel_slots`: `separation_override_enabled`, `separation_artist_minutes`, `separation_title_minutes`. SeparationRulesChecker resolves: slot level → wheel level → template level → daypart level (most specific wins). Slot row in `Entries.vue` gets an expand arrow revealing override fields — hidden by default for clean UI. Recommended: Power Current = 120 min artist sep, Gold = 60 min, Overnight = 30 min.

C3 — Smart Shuffle for Standard Playlists [MEDIUM · Small]

Standard AzuraCast shuffle is purely random. Smart Shuffle adds lightweight artist separation to ANY standard playlist. Useful for background music, specialty shows, and overnight AutoDJ playlists not in a clock wheel. New PlaylistOrders enum case: SmartShuffle. In AutoDJ candidate selection: apply configurable artist separation — no artist repeat within N songs back. No new database tables — reads from recent `song_history` for that playlist. Playlist edit form: Order dropdown → new "Smart Shuffle" option. Configurable per playlist: "No artist repeat within N songs" — default 5, max 20.

Additional Playlist Intelligence

Library Aging System

Tracks not played in X days get an automatic rotation priority bump. Prevents deep catalog from sitting permanently untouched while the same 50 songs cycle. Configurable aging threshold per playlist (e.g. bump priority after 14 days unplayed). Aging status visible in media manager with a "last played" column.

Do-Not-Play List

Temporary or permanent ban on specific songs without deleting from library. Common on-air operational need: pulled for rights issues, out of season, overplayed. Expiry date option: ban automatically lifts after X days. DNP status visible in media manager and shown in clock wheel preview if a slot would select a banned track.

Playlist Health Monitor

Alert when a playlist runs critically low on unplayed tracks before it starts repeating. Configurable threshold: warn when fewer than X tracks remain unplayed. Shown as a warning badge on the Playlists page. Prevents on-air repetition before you notice it manually.

Pending Freelancer Items (from v5)

Feature	Status	Notes
A3 Category/Playlist UI (see Section 2)	Pending	Category + playlist dropdowns on slot rows. Pool validation on save. No backend changes.
C2 Per-slot separation overrides (see Section 4)	Pending	slot > wheel > template > daypart hierarchy. New columns on slot entity.
Wall-clock times in preview modal	Pending	Extend ClockWheelPreviewSimulator to show actual wall-clock times alongside position_seconds. Makes preview readable to non-technical operators.
Schedule boundary warning	Pending	If pinned playlist schedule ends before clock wheel window, warn in preview and log in clock_wheel_events.
Category rotation goals (C2 from freelancer)	Pending	Define per-hour targets: 30% new, 50% recurrent, 20% gold. Clock wheel works toward these goals when selecting candidates.
UI semantics: restrict vs rotate (freelancer)	Pending	Clarify two different slot behaviors with distinct labels + tooltips: "Restrict candidates to this playlist" (filter pool only) vs "Run this playlist rotation rules" (delegate to QueueBuilder, respect shuffle/sequential order). They behave differently under the hood but look identical in the UI today.
CI pipeline wiring	Pending	Wire Codeception unit and functional tests into GitHub Actions. Currently only phpcs/phpstan run in CI.

Enterprise Clock Wheel Additions

D1 — Weekly Program Grid View [LOW · Medium]

New page under Clock Wheels: "Weekly Grid". 7x24 table — columns Mon–Sun, rows hours 0–23, each cell shows daypart name and template name. Click any cell to open the template editor for that hour directly. Color coded by daypart. Empty cells (no template assigned) shown in amber as a visual alert. One-click copy: duplicate a template assignment to another hour or day.

Music Research Score per Track

Add research_score field (0–100) to StationMedia — editable per track in the media manager. In high-listener dayparts, clock wheel automatically weights selection toward higher-scoring tracks. New algorithm option: "Research Score Weighted" alongside existing Random/Oldest/etc. Configurable per wheel: toggle research score weighting on/off.

Sound Code System

Classify songs by sound type: Power, Uptempo, Midtempo, Soft, Ballad, Inspirational, Talk. Per-wheel sound code sequence rules: "after a Ballad, play Uptempo or Midtempo." Enforced in ClockWheelPlaybackPlanner candidate filter — before algorithm selection. Sound code badge shown on each track in media manager.

Reconciliation Log

After each completed wheel hour, compare planned slot sequence against actual clock_wheel_events rows. Flag every discrepancy with the reason (song ran long, emergency override, fallback, etc.). Viewable per wheel per day in the analytics section. CSV export.

Hard Anchor Mode (Exact-Time Enforcement)

Currently position_seconds is a target. Add a hard mode where a slot must fire at its exact second. If the current song would overrun the hard anchor, system fades it out cleanly at cue_out point. Per-slot toggle: "Hard anchor" vs existing "Soft anchor." Hard anchor violations logged in clock_wheel_events with reason and actual drift.

Auto Format Clock Generator

Based on your library categories, daypart goals, and separation rules — suggest a starting clock wheel layout. Operator edits from the suggestion rather than building from zero. Generates a complete wheel with position_seconds, types, and algorithm set.

Live Assist Operator Screen

Stripped-down full-screen view for a live DJ or board op. Shows current wheel, what is queued, countdown timer to next hard anchor. One-click Skip and Hold buttons. No settings access — operations only. Accessible from a separate URL so it can run on a second monitor or tablet. Emergency override toggle shown prominently for breaking news situations.

Wheel Effectiveness Composite Score

One number per wheel per week: drift accuracy % + separation compliance % + fallback rate % + burn rate health %. Shown on clock wheels list page as a weekly score badge with trend arrow.

Clock Wheel Copy / Duplicate

Duplicate a wheel or template as an independent starting point. Export a template to JSON/ZIP importable on another station.

Recent Events List on Live Tab

Show last N slots from clock_wheel_events on the Schedule > Live Clock Wheel tab. Columns: time fired, slot type, track played, drift seconds, fallback reason. Gives operators a post-mortem view of the last hour without opening the analytics modal.

Listener Overlay on Clock Wheel Analytics

Overlay listener count per slot type in the analytics modal. Answers the key question: when my ID slot fires, do listeners drop? Which slot type correlates with tune-outs? Uses existing clock_wheel_events timestamps joined with listener counts at that moment. Shows whether IDs, promos, or talk segments are hurting your audience. No new tables.

Station Health Dashboard

Single page: queue depth, AutoDJ fallback rate, clock wheel drift, AI news status, upcoming schedule conflicts, stream health. Color-coded health indicators per system component. Real-time updating. Accessible from main navigation.

Per-Song Scheduling Restrictions

"Don't play this song before 6am," "Weekends only," "Never during morning drive" — applied per track, not per playlist. Restriction editor in media manager under each track's settings. Respected by both clock wheel planner and general AutoDJ.

Holiday / Special Event One-Time Overrides

Date-specific programming that runs once and automatically reverts to the regular schedule. Example: Christmas Day plays a special playlist, reverts December 26. Managed in a new "Special Events" section on the Schedule page.

Daypart Format Rules in General AutoDJ

Different separation rules, rotation weights, and content behavior for morning/midday/afternoon/overnight in standard AutoDJ. Today this only exists inside clock wheel — general AutoDJ ignores time of day. Extends the existing daypart concept from clock wheels into the standard AutoDJ scheduler.

Scheduled Silence / Sign-Off

Define off-air windows. Plays a sign-off ID, then silence or holding tone until sign-on time. Sign-on plays a wake-up ID and resumes normal programming automatically. Logged in `clock_wheel_events`.

Sweep / Liner Auto-Insertion

Automatic liner/sweeper insertion between songs outside of clock wheel. Configure: play a liner every X songs from a designated liner playlist. Useful for stations not using clock wheels but still wanting bumpers.

Broadcast / Programme Log

Exportable full programme log: what played, when, from which playlist, every day. Separate from SoundExchange — covers all content types. Searchable by date range, content type, playlist, song title.

Stream Quality Monitor

Track bitrate drops, listener connection failures, stream interruptions over time. Visualized as a health chart in Reports. Alert when stream drops below threshold for more than X seconds.

Listener Request Enhancements

Priority queue, per-listener throttle (prevent one person flooding requests), request analytics tab: most-requested songs, peak request times, fulfillment rate. AzuraCast has basic requests today but no analytics or priority system.

Listener Milestone Webhook

Trigger a webhook notification when listener count crosses a threshold. Example: "100 listeners online" posts to social media or Slack. Configurable thresholds: first listener of the day, peak of the week, all-time high.

Listener Geography Alerts

Notify when a new country starts listening for the first time. Useful for your global Christian audience — knowing Germany or Brazil just joined is actionable.

Upcoming Show Embeddable Widget

Embeddable HTML widget showing "what's on next" for your station website. Pulls from schedule API — shows current show/playlist name, next show, time. One-line embed code generated from Public Pages section.

Smart Duplicate Media Detector

Flag songs that exist under different filenames or slightly different metadata. Prevents the same song appearing to play twice in one hour from different playlist sources.

ReplayGain / Audio Normalization Store

Read ReplayGain tags from audio files during media scan, store per track. Pass normalization values to Liquidsoap for level-consistent playback. Eliminates the need to manually set the amplify field per track.

Cue Point Management UI

The metadata fields (cue_in, cue_out, fade_in, fade_out, cross_start_next) already exist in StationMediaMetadata but there is no visual UI to set them per track — currently raw seconds in a form. Build a waveform-style cue point editor: set intro and outro cue points visually so music starts at the beat, not silence, and fades out before trailing silence. Broadcast standard. Ties directly into crossfade intelligence (Section 7) — clean cue points make clean transitions.

Song Attribute Tagging

Tag tracks as: tempo (slow/medium/fast), energy level, era/decade, mood. Foundation for sound codes, rotation goals, and burnout prediction. Tags editable in media manager per track or bulk-set via CSV import.

Why this CAN be built now

The freelancer v5 said to skip crossfade intelligence because it "requires tagging every file with energy/tempo metadata." That objection applies to AUDIO-ANALYSIS based crossfade. We are building CONTENT-TYPE based crossfade — using slot types (Music, Talk, ID, Promo, Ad) that already exist in the system. Zero file tagging needed. Builds directly on existing infrastructure.

What Already Exists

- CrossfadeModes enum: Normal, Smart, Disabled — global station setting
- Per-track metadata: fade_in, fade_out, cue_in, cue_out, cross_start_next, amplify (all in StationMediaMetadata)
- Annotations.php: processAutocueAnnotations() passes all per-track fade data to Liquidsoap
- ConfigWriter.php: writeCrossfadeConfiguration() writes global crossfade settings to Liquidsoap
- Liquidsoap applies azuracast.apply_crossfade(radio) using these annotations
- ClockWheelAnnotator: already applies cue_out for hard anchor enforcement

What to Build

Per-Content-Type Crossfade Profiles

Define different crossfade behavior for each content type transition. Managed in Station Settings > Broadcasting as a matrix. No file tagging — uses the slot type already known at queue time. Examples: • Music → Music: smooth crossfade (existing behavior, configurable duration) • Music → ID: quick fade out, clean cut into ID • ID → Music: cut directly in, no crossfade — IDs should not bleed into songs • Music → Talk: fade out music, brief pause, talk begins clean • Talk → Music: no crossfade — clean cut to music after talk segment • Promo → Music: short crossfade (promo fades while music fades in)

Clock Wheel Slot Boundary Crossfade Rules

When a clock wheel slot changes type, apply the matching content-type crossfade profile automatically. The slot type transition is known at queue time (from ClockWheelPlaybackPlanner) — pass it as a Liquidsoap annotation. No change to the existing annotation path — add a new annotation: clock_wheel_transition_type.

Per-Playlist Crossfade Override

Some playlists (IDs, jingles, station sweepers) should always use a quick-cut profile regardless of global settings. New field on StationPlaylist: crossfade_profile (enum: global/quick_cut/smooth/none). Playlist edit form: new Crossfade Profile dropdown in Advanced tab.

Crossfade Profile Manager UI

New section in Station Settings > Broadcasting: "Crossfade Profiles." Matrix of content type transitions with duration sliders per cell. Preview button: shows how each transition will sound with sample timings. Default profiles pre-set based on content type best practices.

legal_id Quick-Cut Enforcement

When A2 legal_id slot type is built, enforce a clean cut (no crossfade) before and after the legal ID. The ID must fire cleanly at :00 — any crossfade bleeds would push the fire time and defeat the purpose of A2.

Hour Distribution Widget

Show at the top of the clock wheel editor: Music Density X%, Ads/Station ID X%, Talk X%. Calculated from sum of estimated durations per slot type / 3600. Visual progress bars per content type with percentage labels. Updates live as slots are added, moved, or removed. Alert when music density falls below a configurable threshold (e.g. warn if IDs exceed 30% of hour). Computable from existing slot data — quick win, no backend needed.

Smart Weighted Shuffle Algorithm

New algorithm option: "Smart Weighted Shuffle" — combines least-recently-played weight + burn rate health weight + research score weight. Each weight component configurable per wheel (0–100% influence sliders). Single toggle activates all three components together as a smart mode. Logs which weight factor drove the final selection in `clock_wheel_events`. NOT called AI — weighted randomization on existing data, accurately named.

System Integrity / Valid Rotation Indicator

Green badge "Valid Rotation" — no errors or warnings. Yellow badge "Warnings" — pool too small, estimated time under 55 min, duplicate positions. Red badge "Invalid" — empty playlist assigned, `position_seconds` out of range. Shown on clock wheel list page and in editor header. Clicking badge opens a detail panel listing each warning/error with a fix link. Frontend only — backend validation already exists.

Total Estimated Loop Time

Show total estimated loop time as a headline number in the slot editor header. Calculated from sum of slot estimated durations (`duration_seconds` or type defaults). Color-coded: green = 55–60 min (good), yellow = under 55 min (gap risk), red = over 60 min (overrun risk). Updates live as slots are edited. Already partially shown via timeline bar — make it prominent.

Force Reload / Deploy Button

Button on clock wheel list page: "Force Reload" — pushes current config to Liquidsoap immediately without waiting for restart. Confirmation dialog before executing. Logged in audit table: who triggered it and when. AzuraCast already auto-deploys — this adds a manual override for urgent last-minute changes.

Auto-Adjust Crossfades Based on Rotation Logic

When Smart Weighted Shuffle is active, automatically select the matching crossfade profile based on slot type. ID slot detected: apply quick-cut profile. Music slot: apply smooth profile. Talk slot: apply no-crossfade profile. Ties into the Crossfade Intelligence feature in Section 7. No additional configuration needed per slot.

"Most Played" mixes IDs with music in Best & Worst tab

Add content type filter. IDs and promos can be shown separately or hidden from the music ranking. Currently stationid-past-original ranks above most songs in your Most Played list.

Bot/crawler sessions inflate all listener metrics

Covered in B5. Add exclusion toggle applying to all analytics tabs. Show both numbers when toggle is off.

Song Playback Timeline has no playlist filter

Add playlist dropdown filter. Isolate one playlist's performance over time.

Clock wheel analytics modal caps at 30 days

Extend lookback options: add 60-day and 90-day options to the dropdown.

489-second drift displayed as a problem number

Rename to "Avg Transition Offset" with a tooltip: explains this reflects the gap between target position and first play after a transition — expected behavior from the no-hard-cuts rule, not a bug.

Slot category and playlist dropdowns missing from editor

Covered in A3. DB and API support these fields. Frontend dropdowns not built in Entries.vue.

Schedule conflict warnings only surface at runtime

Add pre-save conflict detection to the schedule creation UI. Show conflicts before saving, not just when the wheel tries to run.

CI pipeline not wired to Codeception tests

44 unit tests + functional Cest not running in CI. Wire into GitHub Actions alongside existing phpcs/phpstan.

Slot playlist_id ignores playlist rotation order

Sequential playlists should respect their defined order within a slot. Currently SQL filter only — rotation not wired.

Weekly program grid requires opening each hour individually

Covered in D1. 7x24 grid shows full week at a glance.

FCC / Regulatory Compliance Report

You are an internet radio station — no FCC obligation. The Clock Wheel Performance tab (B2) and the programme log in Section 6 cover automation accountability sufficiently. A dedicated FCC compliance report would be unused infrastructure.

Sub-Second Synchronization

Internet radio delivery latency (buffering, CDN, listener client variation) makes sub-second precision unmeasurable and unachievable end-to-end. The time spent building this would not improve the listener experience at all.

Multi-Station Live Clock Sharing

Shared live clocks across stations cause subtle cross-station bugs. The correct approach is export/import as fully independent copies — already planned in the clock wheel copy/duplicate feature. Build independent templates, not shared live state.

Numeric Priority Stacks

PR13's `is_emergency` boolean is sufficient and clean. A full numeric priority system (100 = Emergency, 50 = Live DJ, 10 = Clock Wheel) is how WideOrbit handles it — and it is also where they become brittle and hard to reason about. One flag, auditable, sufficient.

VAST / Triton Ad Integration

Enterprise programmatic ad tech. Infrastructure cost and complexity exceeds realistic internet radio revenue. Your Ad slot type in clock wheels handles local/direct ad files sufficiently.

Pre-Roll on Stream Join

Static mount intro already handles this in the current AzuraCast setup. Rebuilding this would duplicate existing functionality with no listener benefit.

Audio-Analysis Based Crossfade (Energy/Tempo Tagging)

Building crossfade behavior based on analyzing the acoustic energy of every file in your library requires pre-processing your entire media collection before a line of crossfade logic can run. Our content-type based crossfade (Section 7) achieves professional-sounding transitions without this requirement. Revisit audio analysis only after the library is fully tagged with other metadata.

v0.29	A1 + A2	Songs finish before :00. Station ID fires at top of every hour. Sounds like a professional broadcast station immediately. This is the single highest-impact release. Do this first.
v0.30	A3 + C2 + C3	Category enforcement dropdowns in slot editor. Per-slot separation overrides. Smart Shuffle on standard playlists. Program directors can now configure everything without touching the database.
v0.31	B1 + B2 + B3 + B4	Four new analytics tabs: listener heatmap, clock wheel performance (station-wide), playlist performance, song dropout. Analytics transformed from foundational to enterprise.
v0.32	B5 + B7 + B8 + C1	Bot filtering gives real listener numbers. Repeat listener tracking shows loyalty. Growth trend shows which hours are building audience. Rotation goals stop songs going unplayed for weeks.
v0.33	B9 + B10 + Crossfade Section 7	Retention curve shows when in sessions listeners leave. Daypart audience report maps listeners to programming blocks. Per-content-type crossfade profiles deployed for professional transitions.
v0.34	Section 5 enterprise clock wheel features	Research scores, sound codes, reconciliation log, hard anchors, live assist screen, wheel effectiveness score, weekly grid view, AzuraWheel UI features.
v0.35	Section 6 new features	Station health dashboard, per-song restrictions, holiday overrides, sweep/liner insertion, programme log, stream quality monitor, listener engagement features.

Eternity Ready Radio — Christian Format Rotation Settings

Power Current (TobyMac, Jordan Feliz, Lincoln Brewster): artist separation 120 min, rotation goal 3 days.

Gold (Chris Tomlin catalog, MercyMe classics): artist separation 60 min, rotation goal 7 days.

ER Praise & Worship: artist separation 90 min, rotation goal 5 days.

Overnight clock: relax separation to 20–30 min — sparse inventory at 3am must never trigger fallbacks.

Your existing Station IDs to load into legal_id pool: stationid-past-original, station i.d. 1 seth, station i.d. 2 adult, eternity ready radio legal id.

International reach insight: you have listeners across 6 continents (US, Canada, Germany, Australia, Brazil, India). Your overnight slot — when most US listeners sleep — has a live audience in Australia, Germany, and Brazil. Use the Daypart Audience report (B10) to program overnight content for these time zones rather than treating it as dead air.

Area	Built	Next	Result
Clock Wheels	PR1–PR13 complete	A1,A2,A3,C2,D1,enterprise adds	Broadcast-grade automation
AI Features	AI News + AI DJ built	Merge AI DJ branch, integrate with clock wheel	Full AI broadcast suite
Analytics	6 existing tabs	B1–B10 + enterprise adds	10/10 program director analytics
Playlist Intelligence	Basic rotation	C1,C2,C3 + aging + DNP	Professional rotation control
Crossfade	Normal/Smart/Disabled global	Content-type profiles + per-playlist	Enterprise-grade transitions
On-Air Timing	Gaps at :00	A1 lookahead + A2 legal_id	Sounds like a real station